Performance Evaluation in Ethernet and Wi-Fi Scenarios

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ABSTRACT

This study evaluates the efficiency of wireless and physical network links by comparing theoretical predictions with experimental results in three scenarios: Ethernet-only, Wi-Fi-only, and a mixed configuration. Utilizing iperf3 and Wireshark, we assessed goodput fluctuations and compared these results to calculated expectations derived from protocol overheads.

1 INTRODUCTION

This report is driven by the need to evaluate and contrast the performance of Ethernet and Wi-Fi within a Local Area Network (LAN). A comprehensive understanding of these technologies is essential for refining network configurations in real-world applications. The primary objectives of this experiment are to measure Transfer Control Protocol (TCP) and User Datagram Protocol (UDP) goodput across both wired and wireless mediums, while examining how protocol overhead and wireless-specific limitations affect throughput. Ultimately, this study offers critical insights into network behavior to inform and improve LAN design strategies.

2 TOOLS AND THEORY

2.1 Goodput estimation

Given the capacity C at the physical layer, the **goodput** G is defined as the speed at which useful data is received at the application layer.

$$G = \frac{\text{useful data}}{\text{transfer time}} = \eta_{protocol} * C$$

2.1.1 Ethernet. Here is a breakdown of the goodput for Ethernet connections using TCP and UDP. In both cases the expression used to compute goodput estimation is:

$$\eta_{\text{Eth}} = \frac{MSS}{MSS + \text{Header}_{\text{L4}} + \text{Header}_{\text{L3}} + \text{Header}_{\text{Eth}}}$$

In the context of Ethernet, **TCP** efficiency is determined using the following expression:

$$\eta_{TCP}^{Eth} = \frac{1460}{1460 + 20 + 20 + 38} \approx 94.9\%$$

This calculation accounts for the total overhead, where the Maximum Segment Size (MSS) is derived by subtracting the header lengths from the Ethernet Maximum Transfer Unit (MTU, 1500 B) given the protocol constraints (TCP header: 20 B, IP header: 20 B) [2, 3]. Using TCP protocol, with both machines connected via Gigabit Ethernet (1 Gbps full duplex), the maximum goodput estimation is:

$$TCP : G \leq \eta_{TCP}^{Eth} \times C_{Eth} = 94.9\% \times 1 \text{ Gbps} = 949 \text{ Mbps}$$

Instead, **UDP** efficiency is determined using the following expression:

$$\eta_{UDP}^{Eth} = \frac{1472}{1472 + 20 + 8 + 38} \approx 95.7\%.$$

Efficiency is slightly higher than using TCP because the L4 overhead is smaller (UDP header: 8 B) [1]. Using UDP protocol, with both machines connected via Gigabit Ethernet (1 Gbps), the goodput estimation is:

$$UDP : G \leq \eta_{UDP}^{Eth} \times C_{Eth} = 95.7\% \times 1 \text{ Gbps} = 957 \text{ Mbps}$$

2.1.2 Wi-Fi. Wi-Fi performance is subject to unique constraints that significantly impact overall goodput. The primary limiting factors include the half-duplex nature of the medium and the inherent overhead of the 802.11 protocol, which must manage channel congestion, control frames, and frequent retransmissions.

Under optimal operating conditions, **TCP** efficiency over Wi-Fi typically stabilizes at approximately 70-80% in recent protocol standards (e.g. 802.11ac or subsequent) [5]. This reduction in performance is largely attributed to the protocol's connection-oriented architecture. To ensure reliable data delivery, TCP implements rigorous congestion control and ordered delivery mechanisms. In a wireless half-duplex context, the resulting exchange of control packets and the overhead of managing retransmissions create a performance ceiling that prevents TCP from utilizing the full theoretical bandwidth of the wireless link.

$$\eta_{TCP}^{Wi-Fi(802.11ac)} \approx 70\% - 80\%$$

$$\text{TCP: } G \leq \frac{75\% \times 866.7 \text{ Mbps}}{2} \approx 325 \text{ Mbps}$$

In contrast, **UDP** over Wi-Fi demonstrates higher efficiency, generally ranging between 85% and 90%. This performance advantage derives from UDP's connectionless design, which bypasses the complex handshake and flow required by TCP. By eliminating the necessity for delivery ACKs and flow control, the protocol reduces noise on the network. This results in a more streamlined and rapid transmission process, allowing a larger percentage of the physical layer rate to be dedicated to actual application data.

$$\eta_{UDP}^{Wi-Fi(802.11ac)} \approx 70\% - 80\%$$

$$\text{UDP: } G \leq \frac{85\% \times 866.7 \text{ Mbps}}{2} \approx 368.35 \text{ Mbps}$$

2.1.3 Mixed scenario. In this configuration, the bottleneck is the Wi-Fi link (given 802.11ac protocol usage vs. Gigabit Ethernet). In this specific case, the throughput is not halved because the message is transmitted only once across the Wi-Fi connection.

$$TCP : G \leq \eta_{TCP}^{Wi-Fi} \times C_{WiFi} = 75\% \times 866.7\text{Mbps} = 650.02 \text{ Mbps}$$

$$UDP : G \leq \eta_{UDP}^{Wi-Fi} \times C_{WiFi} = 85\% \times 866.7\text{Mbps} = 736.69 \text{ Mbps}$$

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https://doi.org/

3 LAB SETUP AND SCENARIOS

The following section describes the experimental environment, including the network settings and the specific hardware and software utilized.

3.1 Hardware

The hardware utilized for the experiments consists of:

- **Host A**: Lenovo ThinkPad T490
  Operating System: Kali Linux Live 2026.1 (via SanDisk Extreme Portable SSD)
  Ethernet Interface: Intel Ethernet Connection I219-LN GbE Interface
  Wireless Interface: Intel Canon Point-LP CNVi 802.11ac Wireless Network Controller
- **Host B**: Dell Inspiron 15
  Operating System: Linux Mint 22.3
  Ethernet Interface: Hama USB-A to GbE Adapter Interface
  Wireless Interface: Realtek RTL8852BE PCIe 802.11ax Wireless Network Controller
- **Router**: FRITZ!Box 7530 [4]
  Ethernet ports: 3 GbE ports available
  Wireless Interface: 802.11ac, 2 x 2 MIMO, 80 MHz channel width, 256-QAM

The experimental network configurations are detailed below:

- **Ethernet**:
  - *Protocol:* IEEE 802.3 [7]
  - *Cables*: Cat. 6 UTP, 5GBASE-T, 10GBASE-T
  - *Nominal Speed:* 1 Gbps
  - ***Host IP A*:** 192.168.178.81
  - ***Host IP B*:** 192.168.178.79
  - *Default gateway:* 192.168.178.1
- **Wi-Fi**:
  - *Protocol:* 802.11ac [6]
  - *Security Protocol:* WPA2
  - *Nominal Speed:* 866.7 Mbps
  - *Frequency:* 5 GHz
  - *Bandwidth:* 80 MHz
  - *Channel:* 60
  - ***Host IP A*:** 192.168.178.83
  - ***Host IP B*:** 192.168.178.62
  - *Default Gateway:* 192.168.178.1
- **Mixed**:
  - *Configuration:* Host A connected via Ethernet, Host B connected via Wi-Fi
  - *Nominal Speed:* 866.7 Mbps
  - *Protocol:* IEEE 802.3, IEEE 802.11ac
  - ***Host IP A*:** 192.168.178.81
  - ***Host IP B*:** 192.168.178.62
  - *Default Gateway:* 192.168.178.1

3.2 Scenarios

The evaluation considers the following test scenarios:

- **Both Ethernet:** Host A and Host B establish a connection via a central router by means of Ethernet cables.
- **Both Wi-Fi:** Both Host A and Host B are linked to the same wireless channel.
- **Mixed:** Host A uses a wired Ethernet connection to the router, whereas Host B connects wirelessly.

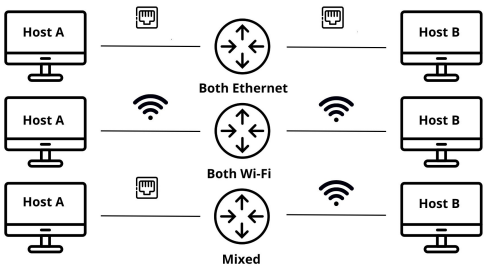


Figure 1: Setup scenarios

4 RESULTS

In this section, the experimental results are compared against the theoretical predictions. Furthermore, a series of relevant plots are analyzed to provide useful insights. In the Ethernet scenario the measured values align almost perfectly with the predicted figures. On the other hand, in both the Wi-Fi and in the mixed scenarios the results are slightly lower than the predictions. The theoretical predictions are compared against the empirical results for each scenario in Fig. 2 and in Fig. 8. These values were generated using the script detailed in Appendix A.

4.1 TCP

Scenario		TCP Goodput Results [Mbps]				
		Mean prediction (Maximum)	*Mean measurement*	*Minimum*	*Maximum*	*Standard Deviation*
Both Ethernet	A → B	949.28	**934.00**	934.00	934.00	0.00
	B → A	949.28	**941.00**	941.00	941.00	0.00
Both 802.11	A → B	325.01	**181.90**	164.00	199.00	11.38
	B → A	325.01	**184.50**	114.00	216.00	36.95
Mixed Mode	A → B	650.02	**490.90**	445.00	547.00	36.92
	B → A	650.02	**330.00**	263.00	402.00	38.26

Figure 2: TCP results

- **Both Ethernet Scenario:** A comparison of the data presented in the table reveals that the measured goodput closely aligns with the theoretical predictions. Furthermore, the standard deviation of the measurements is negligible, which underscores the high reliability and stability of the Ethernet channel. The packet flow is linear and consistent during the test, as shown in Fig. 3. Data loss is negligible, mainly because the channel is reliable.

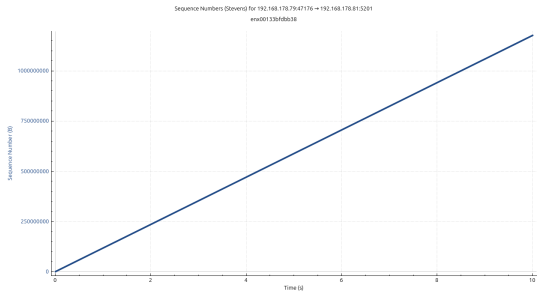
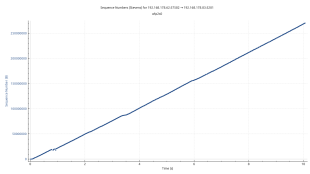
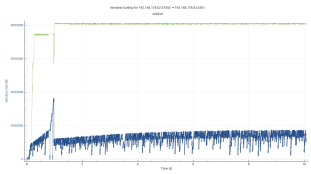


Figure 3: Both Ethernet TCP, Packets sequence

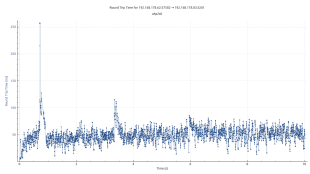
- **Both Wi-Fi Scenario:** To perform the test, the router is configured to transmit and receive on a single channel in the 5 GHz band (channel 60). TCP flow has some interruption around 1*s* as shown in Fig. 5a.



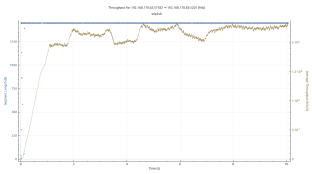
(a) Both Wi-Fi TCP, Packets sequence



(b) Both Wi-Fi TCP, Window



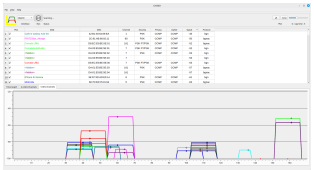
(c) Both Wi-Fi TCP, Round trip Time



(d) Both Wi-Fi TCP, Wi-Fi throughput

Figure 4: Wi-Fi TCP Results

Furthermore, as shown in Fig. 4d, throughput is pretty constant after the initial packet loss. A sudden reduction in window size is shown in Fig. 4b: this is consequent to packet loss. These discrepancies and low performances are primarily attributed to the extremely noisy environment in which the tests were performed, as shown in Fig. 5. As a consequence, measured average goodput is lower than the expected one (184.50 vs 303.34) because the channel is crowded.

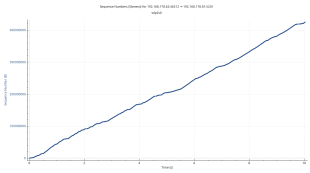


(a) Both Wi-Fi TCP, Wi-Fi Bands Overlapping Condition

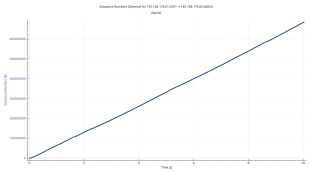
(b) Wi-Fi, Wi-Fi Channels (FRITZ!Box)

Figure 5: Wi-Fi Working Environment

- **Mixed Scenario:** A comparison between the standard scenario (Fig. 11a) and the reverse scenario (Fig. 11b) reveals significant fluctuations in the throughput slope of the forward path, which can be attributed to intermittent packet loss. A comparison of the plots in Fig. 7, specifically for the scenario where the wireless channel is established between Host B and the router, reveals a significant disparity in performance. When Host B transmits to Host A, a high rate of packet loss is observed. Conversely, transmissions from A to B demonstrate greater reliability, which can be attributed to the inherent stability of the Ethernet connection on the source side. The high standard deviation observed is primarily attributable to TCP congestion control mechanisms. Given that Host A is connected to the router via Ethernet, the transmission from A to B exhibits higher speeds, as the wired segment minimizes packet loss and latency before the data enters the wireless bottleneck.

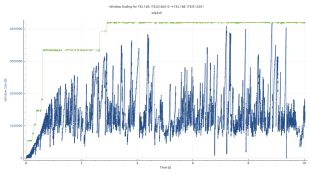


(a) Sequence

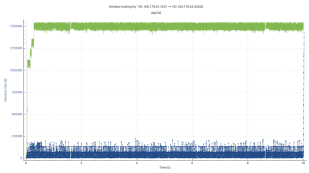


(b) Sequence Reverse

Figure 6: Mixed Scenario TCP



(a) Window



(b) Window reverse

Figure 7: Mixed Scenario TCP

4.2 UDP

Scenario		UDP Goodput Results [Mbps]				
		Mean prediction (Maximum)	*Mean measurement*	*Minimum*	*Maximum*	*Standard Deviation*
Both Ethernet	A → B	957.09	**950.00**	950.00	950.00	0.00
	B → A	957.09	**956.90**	956.00	957.00	0.32
Both 802.11	A → B	368.35	**165.40**	137.00	181.00	14.11
	B → A	368.35	**195.80**	168.00	216.00	13.91
Mixed Mode	A → B	736.69	**489.10**	459.00	555.00	28.57
	B → A	736.69	**389.50**	335.00	443.00	29.12

Figure 8: UDP results

- **Both Ethernet Scenario:** As with TCP, the transmission is effective and the I/0 flow is pretty constant, as shown in Fig. 9.

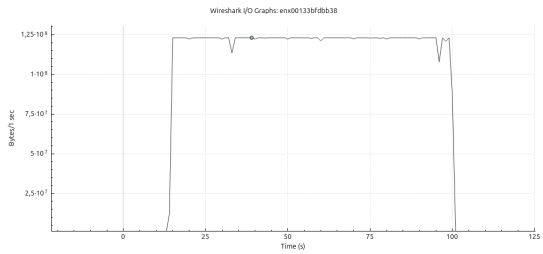


Figure 9: Both Ethernet UDP, Stream

- **Both Wi-Fi Scenario:** In order to show throughput fluctuations during multiple tests, an idle time interval of 2s was introduced. During the measurements, at certain intervals, the throughput exhibits sudden drops within the transmission windows, as shown in Fig. 10.

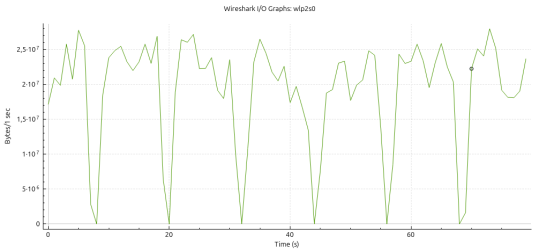
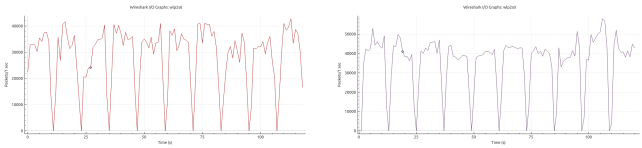


Figure 10: Both Wi-Fi UDP, Stream Reversed

- **Mixed Scenario:** While the throughput profiles for both the direct and reversed cases are qualitatively similar, the reversed scenario exhibits a consistently higher throughput, taking into account the same considerations given in the TCP Mixed Mode scenario.



(a) Stream **(b)** Stream Reversed

Figure 11: Mixed Scenario UDP

5 CONCLUSION

The goal of this study was to analyze the performance of Ethernet and Wi-Fi within LAN settings, focusing on the impact of link capacity and protocol distinctions. Several configurations were examined during the experimental process: Both Ethernet, Both Wi-Fi, Mixed. Considering both Ethernet connection measured data are consistent with the predicted ones. Instead, the theoretical goodput values for WiFi connections, calculated from the overhead impact due to Network protocols, highlights some differences with respect to the actual measurements. As shown in Fig. 5, the wireless environment was crowded and noisy, leading to interference, packet loss and goodput limitations. Future investigations should be performed in a shielded and controlled wireless environment, in order to obtain the best possible results in terms of bitrate and throughput.

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A APPENDIX

The following code is the Python script used to automate the tests in the TCP scenario:

```python
import subprocess, statistics, time

SERVER = # Insert the server IP target
RUNS = 10
SIZE = '1460'

def run_test():
    cmd = ['iperf3', '-c', SERVER, '-l', SIZE]  # -R was added to perform reverse tests
    proc = subprocess.run(cmd, capture_output = True, text = True)

    # Look for receiver line in iperf3 output.
    # Example line: [ 5] 0.00-10.00 sec  1.09 GBytes  936 Mbits/sec  receiver

    for line in proc.stdout.splitlines():
        if 'receiver' in line:
            parts = line.split()
            return float(parts[6])
            # Bitrate is the 7th element of the split line

results = []
print(f'Testing {SERVER} for {RUNS} iterations...')
for i in range(1, RUNS + 1):
    val = run_test()
    if val:
        print(f'Run {i}: {val} Mbps.')
        results.append(val)
    else:
        print(f'Run {i} failed.')

if results:
    print('-' * 20)
    print(f'Minimum bitrate: {min(results):.2f} Mbps')
    print(f'Maximum bitrate: {max(results):.2f} Mbps')
    print(f'Mean bitrate: {statistics.mean(results):.2f} Mbps')
    print(f'Standard deviation: {statistics.stdev(results):.2f} Mbps')
    print('-' * 20)
```

In the UDP scenario, these modifications are applied to the original script:

```python
SIZE = '1472'
cmd = ['iperf3', '-c', SERVER, '-l', SIZE, '-u', '-b', '10G']
# -R was added to perform reverse tests
time.sleep(2) # Added at the end of each 10 s test in W-Fi and Mixed configurations.
```